Lachlan Ewart

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My GitHub
My LinkedIn
My LessWrong
My Website

EDUCATION

University Of Warwick
Bachelor of Science in Mathematics, Current Grade: 1st *2024-2027*

Royal Grammar School Newcastle
*A-Levels in Mathematics, Further Mathematics, Physics, Computer Science [4 A*s]* *2021-2023*

PROJECTS AND EXPERIENCE

Hackathon: Testing Monitoring Models with Encoded Transcripts *Project Writeup*
- Stress-tested CoT monitorability by using transcripts of malicious activities with different encodings on Claude Haiku 4.5.
- Used the proportion of the transcript encoded (in ROT13) as an independent variable to assess what affects the monitoring model's effectiveness.
- Found that partially-encoded reasoning traces evaded a CoT monitor more effectively than fully-encoded ones, suggesting that monitoring models may be weak against less-orthodox encodings.

LLM Continuous Probability Inference *github-repo*
- Studied the low probability estimation (LPE) problem, which concerns efficiently judging the probability of a rare output from an AI in order to prevent extreme cases in deployment.
- In the context of the LPE problem, applied a new MCMC method from literature called COLD (Energy-based Constrained Text Generation with Langevin Dynamics).
- COLD performed 2nd best (after ITGIS), after significantly more compute time. We concluded it was not optimal, but was worth considering in future projects.

Writing *LessWrong Post*
- Made the case for studying how long different monitoring methods stay effective for under RL.
- Proposed experiments to see how adversarial optimisation is affected by monitoring approach.
- Frontpaged, >10 upvotes.

PauseAI UK *PauseAI UK Blogs*
- Wrote (the first two) blogs for PauseAI UK.
- Promoted advocacy and impactful action like emailing your MP.
- Gave digestible descriptions of AI safety ideas, like the Waluigi Effect.

AI Safety Officer Effective Altruism Warwick
- Helped organise and run an AI discussion and up-skilling group.
- Organised a team for hackathons, and arranged speaker events from notable alumni (e.g. Anthropic).

Neural Network from Scratch
- Used an architecture with Pytorch to train an AI on a names data set to generate new plausible names.
- Learned about the general structure and logic behind Neural Networks, and compared different activation functions like ReLU vs GELU.

SKILLS

Programming & Mathematics
- **Machine Learning** - PyTorch, einops, TransformerLens, vLLM, HuggingFace, LoRA finetuning
- **Mathematics** - Linear Algebra, Stochastic processes, Probability theory, Combinatorics, Markov Chains, MCMC Methods